Solent University

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Crime in UK

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1. Introduction

Crime is an action that imitates a notorious deed. An immensely popular topic of discussion in any parliamentary meeting around the globe. Considering the fact that it is not any exception for a country like the UK, which is undoubtedly a very popular destination. Today this report is going to transpire the criminal record in some parts of the UK, subject to be known for its strong law enforcement. This topic is specifically chosen to create awareness in public with “the transparency of increasing criminal activity, at the same time bringing criminal justice professional in scrutiny, with the service they contribute to the society” (Walden University 2023).

1.1 Group Project Aims

As a group project different aspects of crime data have been collected from different sources to identify the status of crime across the UK. Various productive data has been found and analysed to rectify the problem the country is facing. Elaborating the group projects as follows:

- Member 1 – Crime rates in the UK.
- Member 2 – Police recorded crime in Northern Ireland.
- Member 3 – Knife crime in the UK.
- Member 4 – Nature of fraud and computer in the UK.
- Member 5 – Crime rates in London.

1.2 Individual Analysis Objectives

The individual report focuses on the northern region of the UK. Namely Northern Ireland. There are different selective criteria that follows to analyse the data to come up with a productive solution.

Following criteria that are the main objective of this report are as follows:

- Identifying the trend in crime that has been followed each year.
- City that has peaked with the crime occurrence, listed in the police record.
- Crime that has been short listed as the topmost encounter in the police record.
- Grouping the cities and crime type to identify the differences and matching between crime trend.

From this report it can be easily identified the problem the UK is facing at present, and factors related can be used to predict the future vulnerabilities. Also, it can be concluded what actions and which type of actions should be taken based on the analysis.

2. Methodology

2.1 Data Collection

The data used for analysis in this report is mainly retrieved from “data.World” which exclusively produced the dataset based on the “Police recorded crime in the Northern Ireland” (data.World 2020). Based on the analysis requirement it was essential to collect a raw data which subsequently should be processed. It was a tough job to find such dataset as everywhere it is a combined and processed data.

2.2 Data Preparation

Initially the dataset carried miscellaneous numeric information in the same column. This data set also consists of various categorical data that give more depth insight of the report. This categorical data is arranged into rows. This dataset is divided into yearly, monthly, city, and selected crime record wise in those cities. Not only that this dataset gives number of each crime recorded in each city, for each month, and for each year.

As mentioned earlier that the dataset contains numeric data in one column, later this data has been filtered to extract individual precise information to carry forward with the analysis process. There was several information, such as number of crimes recorded and the outcome of that specific crime in that month. Similarly, the percentage of the outcome was also produced to help getting a clear understanding of the report. Each individual information was processed to showcase in a different column for each recorded month and the year simultaneously.

The actual dataset contained information from January 2001 to September 2019. The processed dataset contains information from January 2016 to September 2019. These four years were deliberately chosen to evaluate more detailed information of the trend. Accordingly, the months records for each crime were summed together to give information in a total annual year record for individual city and individual crime, with their outcome.

2.3 Data Analysis

In this project two different analytics have been applied, namely descriptive analytics and predictive analytics.

2.3.1 Identifying the trend in crime followed in an annual year.

To identify the trend or pattern in crime that has been followed over the past few years, descriptive analytical technique has been used. To be more specific measure of association technique applied to find the correlation between the 'number of cases reported' to 'number of cases solved' for each individual type of crime and in individual cities. Using the excel formula 'CORREL' it can be concluded that the scatter plot graph produced, shows a positive correlation of 0.96218. Which also shows a linear correlation.

2.3.2 City with the top recorded crime.

In the further studies to identify the city of most recorded crime, from predictive analytical techniques, time series analysis technique has been applied to scan the data to produce a line graph that give in detail idea of the pattern that has been followed. From there it has been found that Belfast city is the top spotted crime city, with significantly less outcome over the last four years.

2.3.3 Crime type shortlisted as topmost.

Once again from the predictive analytical technique, time series analysis technique applied to run the scanning to locate the most recorded crime pattern over the year. The result found using the line graph, was two different crimes, first 'violence without injury (including harassment)' and second 'criminal damage' was recorded in the top.

2.3.4 Grouping the cities and crime type.

In the final part from descriptive analytics, clustering analysis technique used to group cities and crimes individually. Choosing 3 random values to calculate k values and locate the centralized value that matches with other closely related values. It can be evaluated that, for cities 'Mid Ulster' and 'Mid & East Antrim' was the common centroid. For Crime category, 'Theft vehicle offences' and 'Theft domestic burglary' was the centroids. To calculate all the numeric values needed to standardize using the standardize formula in excel. Then finding the distance from the 3 each centralized values for each value in the data. Afterwards finding the minimum distance and finally locating the name.

2.4 Results

The first analysis derives the trend in the crime followed, showcasing the correlation between Count of crime recorded in the x-axis to the Outcome number in the y-axis.

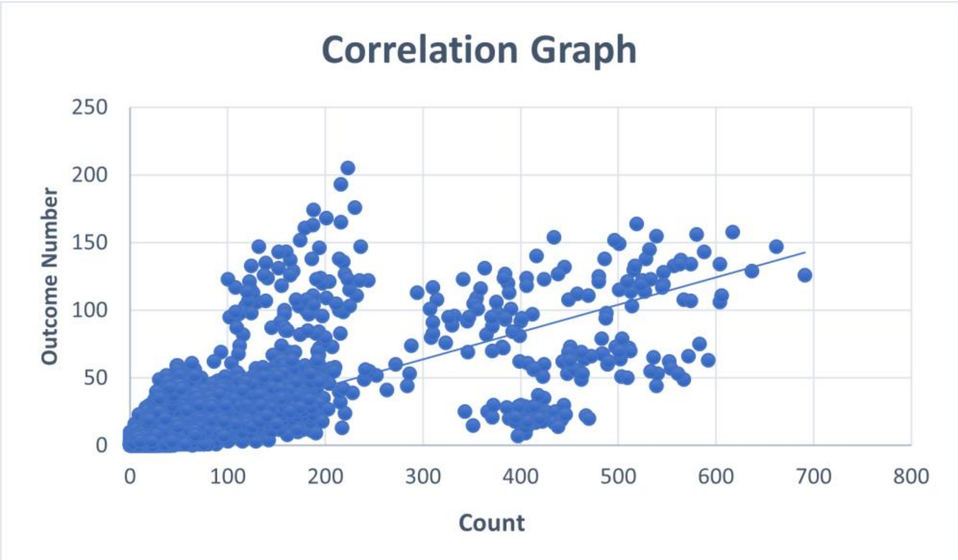


Fig.1: Showing a scatter plot with a positive linear correlation

The second analysis portrays the result of the top listed city in figure 2 and top listed crime in figure 3, for high record in crime rate. As shown blue line shows Count of cases recorded and orange line shows Outcome for each year. Belfast city as shown significantly recorded on the top list.

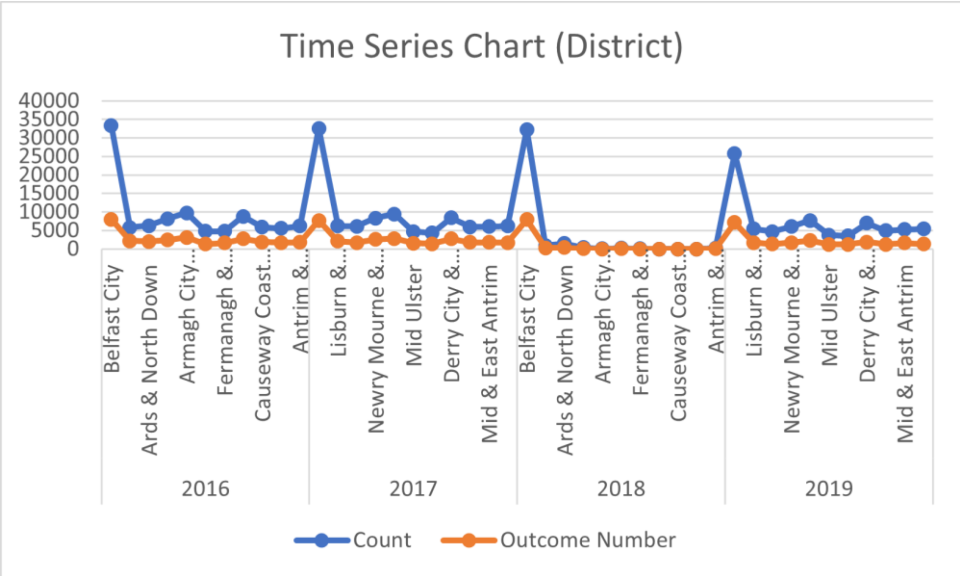


Fig.2: Line graph shows time series analysis based on city

For the graph showing time series based on crime, two different crimes recorded on the top are 'violence without injury (including harassment) and theft domestic burglary' in each consecutive years.

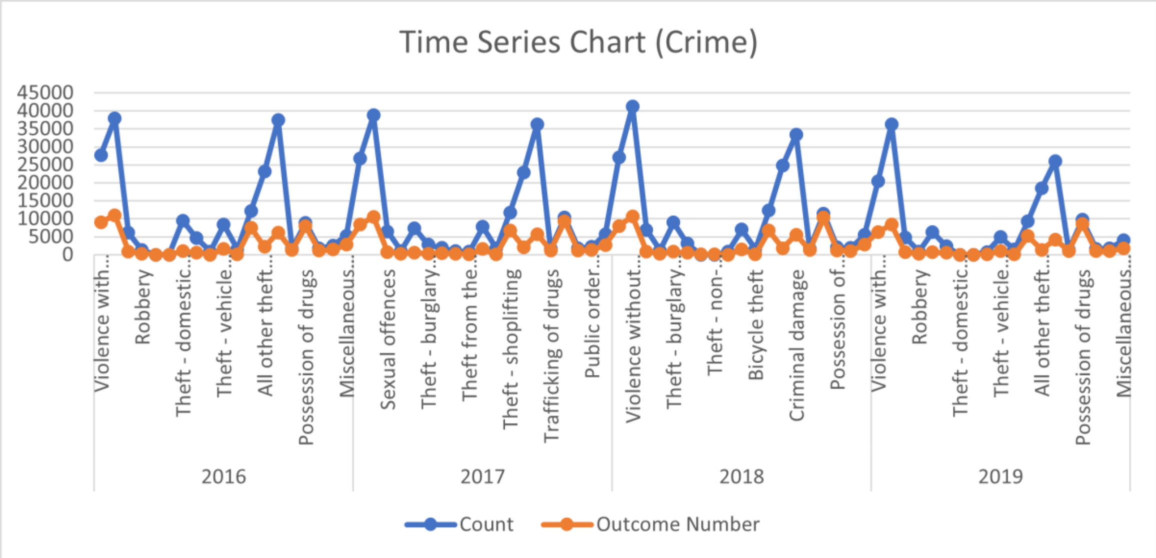


Fig.3: Line graph shows time series analysis based on crime type

The chart below shows the K-means value for clustering analysis. Three different cities chosen for each year as shown in the graph.

2016	**Clustering cities**			
	Cluster center and standardized values			
		Column offset:	5	6
	City	City Index	Count	Outcome Number
	Belfast City	1	2.956010239	2.896121535
	Mid Ulster	6	-0.519896921	-0.68827866
	Mid & East /	10	-0.422740846	-0.509434599
2017	**Clustering cities**			
	Cluster center and standardized values			
		Column offset:	5	6
	City	City Index	Count	Outcome Number
	Belfast City	1	2.957191738	2.896121535
	Mid Ulster	6	-0.528217186	-0.555176039
	Mid & East /	10	-0.358818817	-0.400937893
2018	**Clustering cities**			
	Cluster center and standardized values			
		Column offset:	5	6
	City	City Index	Count	Outcome Number
	Belfast City	1	3.171728783	3.170791855
	Mid Ulster	6	-0.300492852	-0.313182712
	Mid & East /	10	-0.318340165	-0.31977449
2019	**Clustering cities**			
	Cluster center and standardized values			
		Column offset:	5	6
	Belfast City	1	2.956328621	2.961047682
	Mid Ulster	6	-0.548130845	-0.462396356
	Mid & East /	10	-0.312050322	-0.25898872

Fig.4: Clustering analysis K-means centroid values chosen for the city

	Clustering crime type			
	Cluster center and standardized values			
		Column offset:	5	6
2016	Crime_Type	Crime Index	Count	Outcome Number
	Sexual offences	3	-0.322036112	-0.574280612
	Theft - domestic burglary	7	-0.047642432	-0.54677758
	Theft - vehicle offences	10	-0.134224542	-0.342347829
	Clustering crime type			
	Cluster center and standardized values			
		Column offset:	5	6
2017	Sexual offences	3	-0.288062711	-0.596626582
	Theft - domestic burglary	7	-0.664841284	-0.683213085
	Theft - vehicle offences	10	-0.179468624	-0.349566425
	Clustering crime type			
	Cluster center and standardized values			
		Column offset:	5	6
2018	Crime_Type	Crime Index	Count	Outcome Number
	Sexual offences	3	-0.250314163	-0.537284544
	Theft - domestic burglary	7	-0.812101955	-0.781379657
	Theft - vehicle offences	10	-0.247729002	-0.37699167
	Clustering crime type			
	Cluster center and standardized values			
		Column offset:	5	6
2019	Crime_Type	Crime Index	Count	Outcome Number
	Sexual offences	3	-0.302409924	-0.523017527
	Theft - domestic burglary	7	-0.778113253	-0.781983421
	Theft - vehicle offences	10	-0.296807633	-0.407414028

Fig.5: Clustering analysis K-means centroid values chosen for the crime type

Picture below shows the detail calculation of the clustering analysis for cities based on three different cities chosen.

					Standardized		Distance to center				Assigned to		
Year	Index	Policing_District	Count	Outcome Nu	Count	Outcome Nu	To 1	To 2	To 3	Minimum	Index	Center	
	1	Belfast City	33408	8009	2.956010239	2.896121535	0	4.992980607	4.797267084	0	1	Belfast City	
	2	Lisburn & Castlereagh City	5833	2159	-0.38865738	-0.245733596	4.588905658	0.461595008	0.265894531	0.265894531	3	Mid Ulster	
	3	Ards & North Down	6266	2011	-0.3361373	-0.325219845	4.606004313	0.406914365	0.203556501	0.203556501	3	Mid Ulster	
	4	Newry Mourne & Down	8161	2423	-0.10628617	-0.103947313	4.286965403	0.715902913	0.51435737	0.51435737	3	Mid Ulster	
	5	Armagh City Banbridge & Craig	9748	3103	0.086206582	0.261259779	3.895929864	1.126492212	0.923578534	0.923578534	3	Mid Ulster	
2016	6	Mid Ulster	4751	1335	-0.51989692	-0.68827866	4.992980607	0	0.203530099	0	2		0
	7	Fermanagh & Omagh	4803	1643	-0.51358966	-0.52286133	4.871095082	0.165537532	0.091835636	0.091835636	3	Mid Ulster	
	8	Derry City & Strabane	8813	2765	-0.02720282	0.079730372	4.102635657	0.912461151	0.709623628	0.709623628	3	Mid Ulster	
	9	Causeway Coast & Glens	5859	1846	-0.38550375	-0.413836272	4.703353741	0.305581657	0.102594549	0.102594549	3	Mid Ulster	
	10	Mid & East Antrim	5552	1668	-0.42274085	-0.509434599	4.797267084	0.203530099	0	0	3	Mid Ulster	
	11	Antrim & Newtownabbey	6216	1820	-0.34220198	-0.427800072	4.682591022	0.315316646	0.114676524	0.114676524	3	Mid Ulster	
	1	Belfast City	32613	7719	2.957191738	2.901211387	0.005089852	4.908634169	4.679755962	0.005089852	1	Belfast City	
	2	Lisburn & Castlereagh City	6180	2101	-0.34495214	-0.23831724	4.552895848	0.366040403	0.163210789	0.163210789	3	Mid & East Antrim	
	3	Ards & North Down	6140	1722	-0.34994914	-0.450115273	4.7047297	0.206923322	0.04997085	0.04997085	3	Mid & East Antrim	
	4	Newry Mourne & Down	8320	2541	-0.07761253	0.007569659	4.189721813	0.720921132	0.495938905	0.495938905	3	Mid & East Antrim	
	5	Armagh City Banbridge & Craig	9451	2828	0.063677706	0.167954977	3.976847524	0.93448276	0.708620074	0.708620074	3	Mid & East Antrim	
2017	6	Mid Ulster	4713	1534	-0.52821719	-0.555176039	4.90505151	0	0.229096515	0	2	Mid Ulster	
	7	Fermanagh & Omagh	4311	1394	-0.57843706	-0.633412779	4.995826603	0.09296786	0.319807354	0.09296786	2	Mid Ulster	
	8	Derry City & Strabane	8400	2722	-0.06761852	0.108718587	4.113282425	0.808026734	0.586981549	0.586981549	3	Mid & East Antrim	
	9	Causeway Coast & Glens	5923	1765	-0.37705787	-0.426085417	4.706833278	0.198780095	0.031065434	0.031065434	3	Mid & East Antrim	
	10	Mid & East Antrim	6069	1810	-0.35881882	-0.400937893	4.67616583	0.229096515	0	0	3	Mid & East Antrim	
	11	Antrim & Newtownabbey	6234	1666	-0.33820618	-0.481409969	4.718831055	0.203827413	0.083070065	0.083070065	3	Mid & East Antrim	
	1	Belfast City	32280	7943	3.171728783	3.170791855	0	4.918780527	4.936054565	0	1	Belfast City	
	2	Lisburn & Castlereagh City	901	308	-0.22239734	-0.184423309	4.772584304	0.150591809	0.165906503	0.150591809	2	Mid Ulster	
	3	Ards & North Down	1453	351	-0.16268997	-0.165526878	4.716923902	0.201969998	0.219133086	0.201969998	2	Mid Ulster	
	4	Newry Mourne & Down	358	51	-0.28113122	-0.297362444	4.893908097	0.025003071	0.043437371	0.025003071	2	Mid Ulster	
	5	Armagh City Banbridge & Craig	40	14	-0.31552786	-0.313622164	4.929716012	0.015041429	0.006764627	0.006764627	3	Mid & East Antrim	
2018	6	Mid Ulster	179	15	-0.30049285	-0.313182712	4.918780527	0	0.019025722	0	2	Mid Ulster	
	7	Fermanagh & Omagh	65	7	-0.31282372	-0.316698327	4.929979101	0.012822243	0.006316163	0.006316163	3	Mid & East Antrim	
	8	Derry City & Strabane	0	2	-0.31985448	-0.318895587	4.93650404	0.020186868	0.001750894	0.001750894	3	Mid & East Antrim	
	9	Causeway Coast & Glens	0	1	-0.31985448	-0.319335038	4.936814705	0.020315605	0.001576793	0.001576793	3	Mid & East Antrim	
	10	Mid & East Antrim	14	0	-0.31834016	-0.31977449	4.936054565	0.019025722	0	0	3	Mid & East Antrim	
	11	Antrim & Newtownabbey	181	40	-0.30027652	-0.302196415	4.910852101	0.010988427	0.02520484	0.010988427	2	Mid Ulster	
	1	Belfast City	25815	7109	2.956328621	2.961047682	0	4.899102492	4.58812983	0	1	Belfast City	
	2	Lisburn & Castlereagh City	5380	1671	-0.29455109	-0.217488886	4.546571633	0.352537034	0.04503842	0.04503842	3	Mid & East Antrim	
	3	Ards & North Down	4666	1312	-0.408137	-0.427326074	4.775008444	0.144319785	0.19383012	0.144319785	2	Mid Ulster	
	4	Newry Mourne & Down	6028	1669	-0.19146472	-0.218657895	4.474274294	0.431994402	0.12715134	0.12715134	3	Mid & East Antrim	
	5	Armagh City Banbridge & Craig	7722	2263	0.078023427	0.128537898	4.038285833	0.860971814	0.549849442	0.549849442	3	Mid & East Antrim	
2019	6	Mid Ulster	3786	1252	-0.54813084	-0.462396356	4.899102492	0	0.311622656	0	2	Mid Ulster	
	7	Fermanagh & Omagh	3532	1194	-0.58853816	-0.496297628	4.951698403	0.052745116	0.364363887	0.052745116	2	Mid Ulster	
	8	Derry City & Strabane	7017	1842	-0.03413073	-0.117538582	4.291915716	0.618969307	0.31184522	0.31184522	3	Mid & East Antrim	
	9	Causeway Coast & Glens	4938	1269	-0.36486618	-0.452459776	4.762600977	0.183533847	0.200550653	0.183533847	2	Mid Ulster	
	10	Mid & East Antrim	5270	1600	-0.31205032	-0.25898872	4.58812983	0.311622656	0	0	3	Mid & East Antrim	
	11	Antrim & Newtownabbey	5393	1293	-0.292483	-0.438431663	4.702258688	0.256768625	0.180506648	0.180506648	3	Mid & East Antrim	

Fig.6: Clustering analysis calculations based on city data

Picture below shows the detail calculation of the clustering analysis for crime type based on three different crime category chosen.

						Standardized		Distance to center				Assigned to	
	Year	Index	Crime_Type	Count	Outcome Numb	Count	Outcome Numb	To 1	To 2	To 3	Minimur	Index	Center
	2016	1	Violence with injury (including homicide & death/serious injury by unlawful driving)	27672	9006	1.43418109	1.731721002	2.898610412	2.718	2.6003	2.6003	3	Theft - vehicle offences
		2	Violence without injury (including harassment)	37913	11041	2.26753391	2.308717595	3.875248578	3.6761	3.5772	3.5772	3	Theft - vehicle offences
		3	Sexual offences	6090	873	-0.32203611	-0.574280612	0	0.2758	0.2984	0	1	Sexual offences
		4	Robbery	1286	283	-0.7129576	-0.741567093	0.425210975	0.6932	0.7031	0.4252	1	Sexual offences
		5	Theft - burglary residential	0	0	-0.81760477	-0.821807897	0.553947698	0.8176	0.8348	0.5539	1	Sexual offences
0		6	Theft - burglary business & community	0	0	-0.81760477	-0.821807897	0.553947698	0.8176	0.8348	0.5539	1	Sexual offences
		7	Theft - domestic burglary	9462	970	-0.04764243	-0.54677758	0.275768578	0	0.222	0	2	Theft - domestic burglary
		8	Theft - non-domestic burglary	4645	543	-0.43962178	-0.667847627	0.150270341	0.4103	0.4463	0.1503	1	Sexual offences
		9	Theft from the person	889	51	-0.74526314	-0.80734754	0.483157644	0.7447	0.7678	0.4832	1	Sexual offences
		10	Theft - vehicle offences	8398	1691	-0.13422454	-0.342347829	0.298439276	0.222	0	0	3	Theft - vehicle offences
		11	Bicycle theft	1555	74	-0.69106795	-0.800826203	0.433021245	0.6918	0.7213	0.433	1	Sexual offences
		12	Theft - shoplifting	12236	7531	0.1780895	1.313504799	1.952910491	1.8739	1.685	1.685	3	Theft - vehicle offences
		13	All other theft offences	23109	2197	1.06287078	-0.198878406	1.434884639	1.1637	1.2057	1.1637	2	Theft - domestic burglary
		14	Criminal damage	37506	6224	2.23441462	0.942922713	2.972767445	2.7253	2.6949	2.6949	3	Theft - vehicle offences
		15	Trafficking of drugs	1685	1306	-0.6804893	-0.451509347	0.378895071	0.64	0.5571	0.3789	1	Sexual offences
		16	Possession of drugs	8868	7891	-0.09597868	1.415577906	2.002657955	1.963	1.7583	1.7583	3	Theft - vehicle offences
		17	Possession of weapons offences	1724	1136	-0.67731571	-0.499710536	0.36302106	0.6314	0.5654	0.363	1	Sexual offences
		18	Public order offences	2543	1430	-0.61067027	-0.416350832	0.329015947	0.5779	0.4822	0.329	1	Sexual offences
		19	Miscellaneous crimes against society	5321	2823	-0.38461284	-0.021384614	0.556425944	0.6242	0.4071	0.4071	3	Theft - vehicle offences
	2017	1	Violence with injury (including homicide & death/serious injury by unlawful driving)	26843	8337	1.39283513	1.59401196	2.761216176	3.0692	2.4999	2.4999	3	Theft - vehicle offences
		2	Violence without injury (including harassment)	38778	10557	2.37769108	2.234752086	3.888823529	4.2156	3.6356	3.6356	3	Theft - vehicle offences
		3	Sexual offences	6473	747	-0.28806271	-0.596626582	0	0.3866	0.2699	0	1	Sexual offences
		4	Robbery	1091	316	-0.73217622	-0.721022525	0.461206201	0.0772	0.6659	0.0772	2	Theft - domestic burglary
		5	Theft - burglary residential	7360	540	-0.21486898	-0.656371269	0.094481483	0.4508	0.3088	0.0945	1	Sexual offences
		6	Theft - burglary business & community	2798	318	-0.59131748	-0.720445282	0.327558427	0.0824	0.5542	0.0824	2	Theft - domestic burglary
		7	Theft - domestic burglary	1907	447	-0.66484128	-0.683213085	0.386599684	0	0.589	0	2	Theft - domestic burglary
		8	Theft - non-domestic burglary	998	251	-0.73985042	-0.739782934	0.473926025	0.0939	0.6829	0.0939	2	Theft - domestic burglary
		9	Theft from the person	834	73	-0.75338342	-0.791157593	0.504346786	0.1396	0.7241	0.1396	2	Theft - domestic burglary
		10	Theft - vehicle offences	7789	1603	-0.17946862	-0.349566425	0.269872927	0.589	0	0	3	Theft - vehicle offences
		11	Bicycle theft	1743	63	-0.67837429	-0.79404381	0.437397631	0.1117	0.6682	0.1117	2	Theft - domestic burglary
		12	Theft - shoplifting	11761	6747	0.14829408	1.135103491	1.785860098	1.9918	1.5204	1.5204	3	Theft - vehicle offences
		13	All other theft offences	22900	2080	1.06746546	-0.211893884	1.40906915	1.7953	1.2545	1.2545	3	Theft - vehicle offences
		14	Criminal damage	36293	5707	2.1726331	0.834936945	2.846822437	3.2181	2.6335	2.6335	3	Theft - vehicle offences
		15	Trafficking of drugs	1614	1233	-0.68901915	-0.456356446	0.424784386	0.2281	0.5206	0.2281	2	Theft - domestic burglary
		16	Possession of drugs	10400	9356	0.03598667	1.88811745	2.505785486	2.6651	2.248	2.248	3	Theft - vehicle offences
		17	Possession of weapons offences	1853	1108	-0.66929727	-0.492434156	0.39521621	0.1908	0.5102	0.1908	2	Theft - domestic burglary
		18	Public order offences	2178	1266	-0.64247882	-0.44683193	0.384771645	0.2374	0.4731	0.2374	2	Theft - domestic burglary
		19	Miscellaneous crimes against society	5701	2720	-0.35176684	-0.02717601	0.573002766	0.7269	0.3655	0.3655	3	Theft - vehicle offences
	2018	1	Violence with injury (including homicide & death/serious injury by unlawful driving)	27007	7985	1.36969311	1.448097366	2.562452908	3.1194	2.4386	2.4386	3	Theft - vehicle offences

Fig.7.1: Clustering analysis calculation based on crime type

	19	Miscellaneous crimes against society	5701	2720	-0.35176684	-0.02717601	0.573002766	0.7269	0.3655	0.3655	3	Theft - vehicle offences
2018	1	Violence with injury (including homicide & death/serious injury by unlawful driving)	27007	7985	1.36969311	1.448097366	2.562452908	3.1194	2.4386	2.4386	3	Theft - vehicle offences
	2	Violence without injury (including harassment)	41232	10709	2.51887795	2.214128574	3.903677637	4.4798	3.7905	3.7905	3	Theft - vehicle offences
	3	Sexual offences	6954	925	-0.25031416	-0.537284544	0	0.6125	0.1603	0	1	Sexual offences
	4	Robbery	1193	272	-0.71572392	-0.72091831	0.500327497	0.1138	0.5808	0.1138	2	Theft - domestic burglary
	5	Theft - burglary residential	8956	870	-0.08858003	-0.5527514	0.162472008	0.7588	0.2371	0.1625	1	Sexual offences
	6	Theft - burglary business & community	3086	508	-0.56279549	-0.654551436	0.333760553	0.2797	0.4199	0.2797	2	Theft - domestic burglary
	7	Theft - domestic burglary	0	57	-0.81210196	-0.781379657	0.612525875	0	0.6943	0	2	Theft - domestic burglary
	8	Theft - non-domestic burglary	0	65	-0.81210196	-0.779129933	0.611632827	0.0022	0.693	0.0022	2	Theft - domestic burglary
	9	Theft from the person	853	45	-0.74319126	-0.784754244	0.551515261	0.069	0.6417	0.069	2	Theft - domestic burglary
	10	Theft - vehicle offences	6986	1495	-0.247729	-0.37699167	0.160313719	0.6943	0	0	3	Theft - vehicle offences
	11	Bicycle theft	1538	87	-0.68785266	-0.77294319	0.496965724	0.1245	0.592	0.1245	2	Theft - domestic burglary
	12	Theft - shoplifting	12313	6696	0.18261951	1.085610499	1.67964874	2.1154	1.5246	1.5246	3	Theft - vehicle offences
	13	All other theft offences	24807	1837	1.1919633	-0.280815946	1.464902868	2.0656	1.4429	1.4429	3	Theft - vehicle offences
	14	Criminal damage	33428	5566	1.88842182	0.767836907	2.505500668	3.1133	2.4236	2.4236	3	Theft - vehicle offences
	15	Trafficking of drugs	1690	1307	-0.67557314	-0.429860197	0.438617361	0.3771	0.4311	0.3771	2	Theft - domestic burglary
	16	Possession of drugs	11464	10309	0.11403196	2.101642347	2.663960066	3.0281	2.5049	2.5049	3	Theft - vehicle offences
	17	Possession of weapons offences	2039	1249	-0.64737873	-0.4461707	0.407384343	0.3735	0.4056	0.3735	2	Theft - domestic burglary
	18	Public order offences	1920	1103	-0.6569923	-0.487228173	0.409747172	0.3325	0.4238	0.3325	2	Theft - domestic burglary
	19	Miscellaneous crimes against society	5531	2791	-0.36527304	-0.012536294	0.537192953	0.8893	0.3829	0.3829	3	Theft - vehicle offences
2019	1	Violence with injury (including homicide & death/serious injury by unlawful driving)	20434	6346	1.23006245	1.44470161	2.494071007	2.9985	2.4003	2.4003	3	Theft - vehicle offences
	2	Violence without injury (including harassment)	36303	8396	2.78976	2.165027365	4.097206387	4.6276	4.018	4.018	3	Theft - vehicle offences
	3	Sexual offences	4842	746	-0.30240992	-0.523017527	0	0.5416	0.1157	0	1	Sexual offences
	4	Robbery	941	264	-0.68582288	-0.692381925	0.41915366	0.1286	0.4822	0.1286	2	Theft - domestic burglary
	5	Theft - burglary residential	6321	657	-0.15704521	-0.554290207	0.148690554	0.6615	0.2027	0.1487	1	Sexual offences
	6	Theft - burglary business & community	2320	509	-0.55028674	-0.606294212	0.261491726	0.2877	0.3222	0.2615	1	Sexual offences
	7	Theft - domestic burglary	2	9	-0.77811325	-0.781983421	0.5416244	0	0.6099	0	2	Theft - domestic burglary
	8	Theft - non-domestic burglary	0	33	-0.77830982	-0.773550339	0.53781726	0.0084	0.6049	0.0084	2	Theft - domestic burglary
	9	Theft from the person	776	63	-0.70204003	-0.763008986	0.466154616	0.0784	0.5391	0.0784	2	Theft - domestic burglary
	10	Theft - vehicle offences	4899	1075	-0.29680763	-0.407414028	0.115739167	0.6099	0	0	3	Theft - vehicle offences
	11	Bicycle theft	1438	76	-0.63697483	-0.758441067	0.40909402	0.1431	0.4888	0.1431	2	Theft - domestic burglary
	12	Theft - shoplifting	9324	5177	0.13810708	1.03394024	1.61807686	2.034	1.5055	1.5055	3	Theft - vehicle offences
	13	All other theft offences	18462	1386	1.03624283	-0.29813534	1.357410475	1.8778	1.3375	1.3375	3	Theft - vehicle offences
	14	Criminal damage	26027	4260	1.779775	0.711726231	2.420761491	2.9621	2.359	2.359	3	Theft - vehicle offences
	15	Trafficking of drugs	1267	1040	-0.6537817	-0.419712273	0.366243227	0.383	0.3572	0.3572	3	Theft - vehicle offences
	16	Possession of drugs	9713	8608	0.17634027	2.23951959	2.803714156	3.1687	2.6889	2.6889	3	Theft - vehicle offences
	17	Possession of weapons offences	1567	979	-0.62429596	-0.441146356	0.332134773	0.3739	0.3292	0.3292	3	Theft - vehicle offences
	18	Public order offences	1803	974	-0.60110051	-0.442903248	0.309248058	0.3825	0.3064	0.3064	3	Theft - vehicle offences
	19	Miscellaneous crimes against society	4019	1857	-0.38329915	-0.132636106	0.398673702	0.76	0.2881	0.2881	3	Theft - vehicle offences

Fig.7.2: Clustering analysis calculation based on crime type

2.5 Conclusion

Criminal activity is a global issue. Therefore, keeping the up-to-date data is essential to keep track of the crime and stop from progression. All the data provided in this report is meant to follow the pattern and rectify the problem causing this rise in crime records. Crime rate increase in some places may cause due to the number of resources civil defence using to stop the crime, number of battalions tackling the geographical range, also the number of forces appointed to keep the process running.

In conclusion, factors that mentioned above could be the reason of drastic increase in crime rates in some places of Northern Ireland. Government must immediately take action to protect civilian from future attacks.

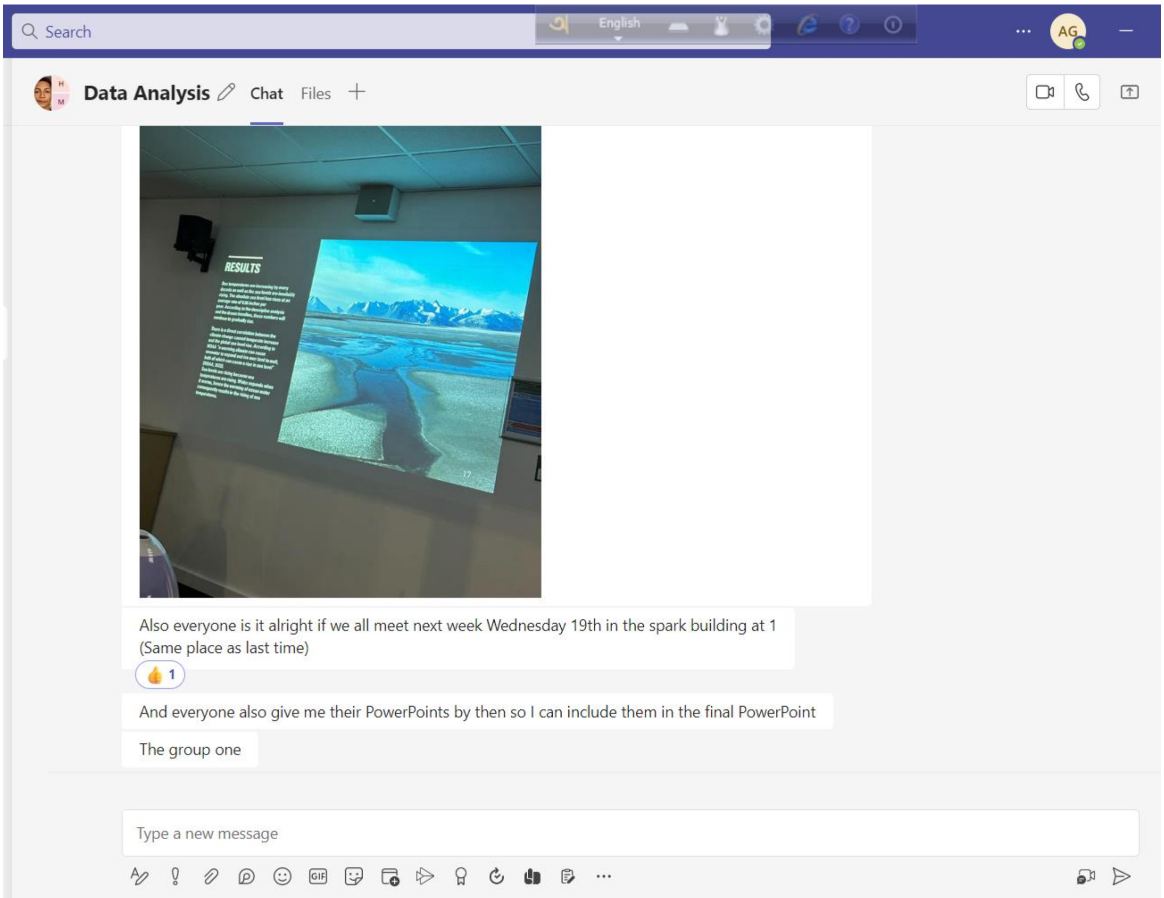
3. Project Management

3.1 Project Plan

As it is a group project. From the beginning of the project the searched individually for a domain. First proposal came up on air pollution in Britain. Later, the group unanimously decided to work on crime as it is one of the most controversial topics around the world. Since everyday there is a significant record in this area. The group decided to work on this domain.

3.2 Project Collaboration

One of the responsible members of the group created teams chat to discuss further regarding the assessment topic. Also, the group arranged meetups on campus to develop the plan and help in person to solve any challenges faced by any member of the group.



3.3 Learning reflection

From the personal viewpoint this project was quiet challenging in the beginning. As it is completely based on a dataset, therefore it was a tough job to find an appropriate dataset which is unprocessed and contains enough information to carry the analysis. Secondly, it was initially confusing to understand what analysis technique can be applied to give more authentic result. I must say pre-recorded videos from the tutor in sol page helped me decide which analysis process I should choose to apply on my project. I certainly feel there are many things to enjoy while working on a dataset. I do realize it requires a lot of brain work but at the same time excel makes it easier in many ways. I do want to explore in this field to enhance my knowledge in this field.

Group work was not that difficult as every member of the group participated accordingly. We communicate through team's chat. Although initially finding a dataset was the most challenging part for every one of us. Nevertheless, we had frequent meetup on campus, where we were able to utilize our time helping each other and solving problems together as a group. Although there were many challenges, but I personally enjoyed working with the group and with my individual project.

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